BDS-SMC2 UAV Rescue System

Publication Plan — Two Papers (Rev. 2)

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Updated 12 June 2026 — post-audit numbers, 112-bit rescue payload, ROS 2 integration

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1. Publication Strategy

Revision 2 consolidates the encoding results (Gaps 1 + 6) into Paper 1: the 112-bit rescue payload is now the carrying contribution and belongs with the reliability and latency evidence that proves it deliverable. Paper 2 becomes the end-to-end system paper (RTK injection, GCS decode, ROS 2, UAV mission pipeline), matching the group integration now in place.

■ Positioning note: the BDS-3 GSMC system paper (Geodesy & Geodynamics, 2021; 2149 TX, 97.72%) already field-measured the link. Our claim is therefore the NARROWED one below — never claim 'first field measurement of BDS-3 SMC'.

Paper	*Scope*	*Target Venue*	*Core Claim*
Paper 1	Gaps 1, 2, 3, 6 + 112-bit rescue payload	MDPI Drones (primary) / Sensors / IEEE Access	First environment-stratified characterisation of BDS-3 SMC carrying a complete 6-field rescue payload: 112 bits, delivery reliability ≥93.7% (Wilson 95% LB), 2.57 s mean latency, integrated into a UAV mission pipeline
Paper 2	End-to-end system: RTK injection, portal GCS, ROS 2, MAVLink	IEEE Access / MDPI Drones	Full survivor-to-UAV pipeline: RTK coordinate → 112-bit BDS-SMC → portal → decode → ROS 2 mission trigger, with stage-by-stage latency budget

2. Paper 1 — Transmission Layer (Gaps 1, 2, 3, 6 + 112-bit payload)

Field	*Detail*
Working title	A 112-bit Binary Rescue Payload over BeiDou-3 Short Message Communication: Environment-Stratified Reliability and Latency for UAV Search-and-Rescue
Venue	MDPI Drones (primary — publishes this genre, fast review); fallbacks: MDPI Sensors, IEEE Access
Submission	After 1 remaining field day (Gap 2 midday + evening, 112-bit verification TX) — target August 2026
Status	Empirical work ~75% complete. Defence sections drafted (BDS_SMC2_Paper1_Sections.md): comparison table, emulated-RTK framing, Wilson claim, exclusion appendix, repeat-TX policy.

2.1 Data Status (audited 2026-06-12)

Dataset	*Requirement*	*Collected*	*Status*

Gap 3 reliability	4 environments x 3 locations x ~20 TX	232 valid TX (61/57/57/57)	**COMPLETE**
Gap 2 latency	3 time-of-day sessions x 30 TX, ALL on the 112-bit payload (Option B)	0 of 90 (30-TX ASCII baseline archived as secondary comparison)	**NEED morning + midday + evening**
Gap 1 / Gap 6 encoding	Bit counts + round-trip verification	Software-verified incl. 112-bit on lab T001-T006	**COMPLETE**
112-bit hardware proof	1 TX accepted by BDS module + portal receipt	0	**NEED 1 test TX**

2.2 Key Results to Report (verified against raw data, 2026-06-12)

Metric	*Value*	*Source*
Delivery success (valid TX)	232/232; report as Wilson 95% LB ≥93.7% per env, ≥98.4% pooled	gap3_field_test.csv
Environment effect	None detectable: chi-sq=0.000, df=3, p=1.000; Fisher pairwise all 1.0	gap3_analysis.py
Mean TX latency (T1→T3)	ASCII baseline: 2574.5 ms (n=30, sd 1094.7) — archived; operational-payload sessions TBD	gap2_latency_ascii_baseline.csv
ANOVA time-of-day	TBD after 3 sessions on 112-bit payload (Option B); ASCII-vs-binary secondary comparison via --ascii-baseline	gap2_analysis.py
Gap 1 encoding	ASCII 264 bits vs 112-bit rescue payload (-57.6%)	decode_binary.py round-trip
Gap 6 telemetry	ASCII 368 / Huffman 184 / Binary 112 bits (-69.6% vs ASCII; 39% under Huffman with MORE fields)	gap6_telemetry.csv
Payload completeness	All 6 lab rescue-report fields (T001-T006) round-trip bit-perfect at 7dp	decode_binary.py test
BDS-3 headroom	98 bits under the 210-bit regional limit	design
Excluded records	33 rows, single cause (T3 instrumentation bug, OS-1, re-collected) — full appendix drafted	Paper1_Sections C.2

2.3 Remaining Task List — Paper 1

#	*Task*	*When*	*Tool / Ref*
1	Flash MODE 1 firmware once; Gap 2 morning session (30 TX) — TX #1 doubles as the 112-bit acceptance check + on-air bit-accounting check on the portal	Field day, 08:00-10:00	run_gap2_morning.bat
2	Gap 2 midday session (30 TX) + optional power measurement	Field day, 12:00-14:00	run_gap2_midday.bat
3	Gap 2 evening session (30 TX)	Field day, after 18:00	run_gap2_evening.bat

4	Activate portal reader (browser tokens) + run dashboard for portal-corroborated delivery evidence	Field day	portal_reader.py / tx_dashboard.py
5	Optional: power draw per TX (USB meter)	Field day	J/message metric
6	Run ANOVA across 3 sessions; regenerate figures	After field day	gap2_analysis.py / generate_figures.py
7	Fill [CITE] placeholders in drafted sections (LoRaWAN, COSPAS-SARSAT, Iridium, BDS-3 GSMC paper)	This week	BDS_SMC2_Paper1_Sections.md
8	Assemble full draft: drafted sections + results + figures	After analysis	venue template
9	Supervisor review of submission-ready draft (not an outline)	Before submission	email
10	Submit	Target Aug 2026	venue portal

3. Paper 2 — End-to-End Rescue Pipeline (system paper)

Field	*Detail*
Working title	End-to-End BDS-3 Short Message Rescue System: RTK Coordinate Injection, GCS Decoding, and UAV Mission Integration via ROS 2
Venue	IEEE Access or MDPI Drones
Prerequisite	Paper 1 submitted. Group integration validated (ROS 2 node + verify_integration.sh already in the team repo).
What it proves	Paper 1 proves the transmission layer; Paper 2 proves the full pipeline: RTK fix → encode → satellite → portal → decode → ROS 2 topic → UAV mission, with a latency budget per stage.
Note	Group stack is ROS 2 (not ROS 1 as in Rev. 1 of this plan) and uXRCE-DDS/px4_msgs (not MAVROS). EmergencyCoordinate.msg extension (alt / uncertainty / priority / survivor_id) pending group approval.

3.1 Data To Collect (after Paper 1)

Dataset	*Minimum sample*	*Purpose*
RTK → BDS TX accuracy chain	30 TX per coordinate set	Coordinate accuracy loss through transmission chain
GCS decode latency	30+ samples	Portal-poll + decode stage of the latency budget
Multi-survivor scenario	3 runs x 6 TX (T001-T006 messages ready)	Survivor-ID field + multi-victim triage validation
ROS 2 publish rate	50+ samples	Real-time capability of the integration layer
Mission-trigger latency	20+ samples	Topic publish → mission reaction (uXRCE-DDS path)
UAV positional error model	Derived from Gap 2 data	No new hardware — modelled from latency distribution

4. Combined Timeline (Rev. 2)

Period	*Activity*	*Output*
Now (June W2)	Fill citations; group meeting on .msg extension; push node to team repo	Drafted sections citation-complete
Field day	Gap 2 midday + evening; 112-bit TX; portal tokens; dashboard evidence	Paper 1 empirical record COMPLETE
June W3-W4	ANOVA, figures, assemble Paper 1 full draft	Submission-ready draft to supervisor
July	Revise per supervisor; Ubuntu integration demo with group	Paper 1 final; integration video/screenshots for Paper 2
August 2026	Submit Paper 1 (MDPI Drones)	Under review
Sep-Oct 2026	Collect Paper 2 datasets; write system paper	Paper 2 draft
Nov 2026	Submit Paper 2 (IEEE Access / Drones)	Under review